

Track 3: Merble| Design Brief

About:

Merble is an AI loan officer platform for banks and credit unions. It monitors member data to detect financial signals and surfaces loan opportunities before the borrower goes looking elsewhere.

The Problem:

Most people do not call their bank when they need a loan. They open Google. By the time they search "best HELOC rates," the bank has already lost them. The problem is that banks actually have the data to see the need coming weeks earlier, such as a job change, rising credit card balances, a tuition bill hitting the account, or a renovation permit filed nearby. Nobody is connecting those signals in time. Merble wants to change that.

Your Task:

Build a system that reads member data, detects who is likely to need a loan in the next 30 to 90 days, and surfaces a ranked list of alerts for a loan officer. The loan officer has two hours in their day. Your system tells them who to call first and why.

Important Points to think about:

- Detect the Signal: Which points actually predict a borrowing need? Job change? Balance spike? A permit filed nearby?
- Classify the product: Does this member need a HELOC, a refi, debt consolidation, an auto loan, or a personal loan?
- Rank the list: Out of all flagged members, who should the loan officer call first? Show your reasoning
- Explain it clearly: A loan officer and a compliance reviewer both need to trust the output. No black boxes

Deliverables:

- A deck (8-12 Slides): Walk through your signal logic, how you classify products, how you rank members, and why a compliance reviewer can trust the output
- A working system: Build something that takes in synthetic member records and outputs a ranked alert list. Use whatever tool fits your team (Python script, Spreadsheet with formulas, Airtable or Notion, Vibe-coded app, Claude-generated code)
- A loan officer alert card: Design the actual notification a loan officer sees when a member is flagged.

SUBMIT NO LATER THAN FRIDAY @ 5:00 PM

Rubric

Signal logic Are the right data points used to predict borrowing need? Member records must include all six fields, plus at least one external signal. (Account balances, Transactions [60 days], Credit score range, Employment Status, Home ownership, Public signals)	Needs work: Missing required fields, no external signals, logic is generic	Good: All six fields are present, at least one external signal, and the reasoning is sound	Excellent: All fields are used meaningfully, external signals are specific and non-obvious, and strong predictive logic	Score: ____/35
PM Tools & Frameworks Did they use structured thinking to understand the customer?	Needs work: No frameworks used. Pure gut feeling	Good: At least one framework is present, used somewhat meaningfully	Excellent: Frameworks drive real insight that shapes the strategy	Score: ____/10
Product classification and ranking Does the system correctly match signals to loan products and produce a defensible ranked list of at least 10 members? (HELOC, Refi, Debt Consolidation, Auto Loan, Personal Loan)	Needs work: Products are assigned arbitrarily, ranking has no clear logic, and fewer than 10 members	Good: Products mostly match signals, ranking makes sense, 10 members present	Excellent: Classification is sharp, ranking is clearly prioritized by urgency and fit, and every member placement is defensible	Score: ____/25
Loan Officer Card Does each alert card include all four required elements? (Member summary & why flagged, Recommended product and signals, suggested outreach line, confidence level)	Needs work: Missing required elements, no visual design, reasoning unclear	Good: All four elements are present, looks like a real UI, mostly explainable	Excellent: All four elements are polished and specific; a loan officer could act on them immediately with full confidence	Score: ____/10
Pitch Is the system logic clear, and could Merble build on this immediately?	Needs work: Hard to follow, too many gaps in the reasoning	Good: Clear walkthrough, most decisions explained	Excellent: Tight, confident, Merble could hand this to an engineer and say build it	Score: ____/20

Total: ____/100